Introduction to Regression

1 Introduction to Regression

What Is Regression?

Regression is a statistical technique for modeling the relationship between a **response** (dependent) variable and one or more **predictor** (independent) variables.

Instead of treating events in a system as isolated, regression lets us ask: *if this variable changes, how does that one respond, on average?*

- Response variable: what we want to explain or predict (denoted y)
- Predictor variable(s): what we use to explain it (denoted x)
- The regression equation summarizes the average relationship between them

Why Regression?

- **Business:** does advertising spend increase sales, and by how much?
- **Medicine:** how does drug dosage affect blood pressure reduction?
- **Engineering:** how does curing temperature affect concrete strength?
- **Economics:** how does income relate to household spending?
- **Environment:** how does fertilizer amount relate to crop yield?

2 Simple Linear Regression (SLR)

The SLR Model Equation

For an outcome variable Y and a single predictor variable X, the model is given as

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)$$

- β_0: the intercept — expected or average value of Y when the value of X is zero
- β_1: the slope — the expected or average change in Y for a one-unit increase X
- ε_i: random error — everything affecting Y that X does not capture
- β_0, β_1 are unknown **population parameters**; we estimate them from data as $\hat{\beta}_0, \hat{\beta}_1$

The SLR Model in Terms of Observed Values

For a single predictor X (advertising spend) and response Y (sales), and observed data points (x_i, y_i), $i = 1, 2, ..., n$, the SLR model is

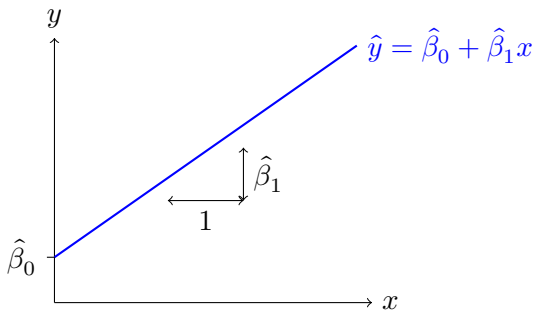
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2), \qquad i = 1, \dots, n.$$

The fitted line is given by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

What are the interpretations of $\hat{\beta}_0$ and $\hat{\beta}_1$ in this context?

Interpretation of the Parameters



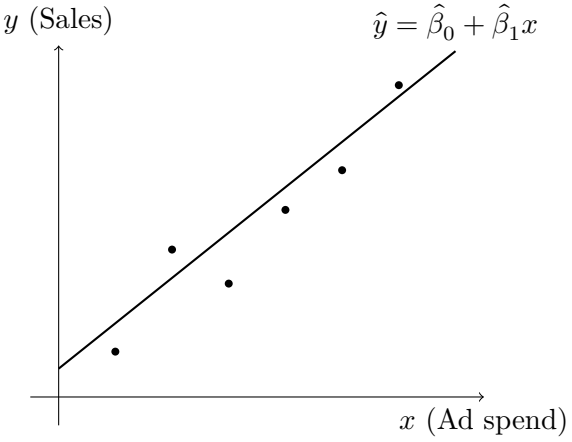
Assumptions of the SLR Model

- **Linearity:** the mean of y is a linear function of x, i.e. $\mathrm{E}[\varepsilon_i] = 0$
- **Independence:** $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$
- **Normality**: $\varepsilon_i \sim N(0, \sigma^2)$
- **Constant variance (homoscedasticity):** $\mathrm{Var}(\varepsilon_i) = \sigma^2$ for all i
- x_i values are fixed, known constants, measured without error

3 Ordinary Least Squares (OLS) Estimation

The Regression Line

We want a line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ that fits the scatter of (spend, sales) points as closely as possible.

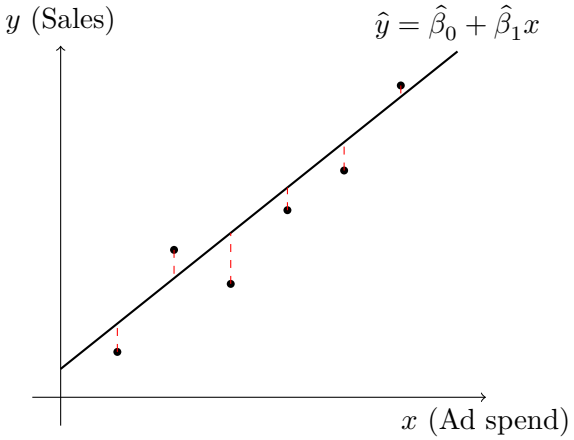


Residuals

For each data point, the **residual** is the vertical gap between the observed y_i and the fitted value $\hat{y}_i$:

$$\hat{\varepsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Ordinary Least Squares (OLS) chooses $\hat{\beta}_0, \hat{\beta}_1$ to make these residuals collectively as small as possible.



Setting Up the Least Squares Problem

OLS minimizes the sum of squared residuals:

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

To minimize $S(\beta_0, \beta_1)$, we take partial derivatives with respect to β_0 and β_1, set them equal to zero, and solve.

Since S is a quadratic (convex) function, this stationary point is the global minimum.

First-Order Conditions

Differentiate $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$:

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i)$$

Setting both equal to zero and evaluating at the minimizing values $\hat{\beta}_0, \hat{\beta}_1$ gives the **normal equations**.

The Normal Equations

$$\sum_{i=1}^{n} y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i y_i = \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2$$

This is a system of two linear equations in two unknowns, $\hat{\beta}_0$ and $\hat{\beta}_1$. We solve it directly using algebra.

Solving for $\hat{\beta}_1$

From the first normal equation, dividing by n:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Substitute into the second normal equation:

$$\sum_i x_i y_i = (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since $\sum_i x_i = n\bar{x}$, this becomes

$$\sum_i x_i y_i - n\bar{x}\bar{y} = \hat{\beta}_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right)$$

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\left(\sum_i x_i^2 - n\bar{x}^2 \right)}$$

Solving for $\hat{\beta}_1$ (cont.)

Define

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y}, \qquad S_{xx} = \sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$$

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

Solving for $\hat{\beta}_0$

Having solved for $\hat{\beta}_1$, we plug back into the first normal equation:

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

Interpretation for our example: $\hat{\beta}_1$ is the estimated extra sales revenue per extra unit of advertising spend, and the fitted line passes through the point of averages $(\bar{x}, \bar{y})$.

OLS Estimators — Summary

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Estimator of β_1 in terms of Covariance

Recall that

$$\text{Cov}(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{S_{xy}}{n-1}$$

and

$$\text{Var}(X) = \frac{\sum_i (x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1},$$

hence,

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Relationship between $\hat{\beta}_1$ and the Correlation Coefficient

Recall that the population correlation coefficient ρ is estimated by

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}},$$

therefore

$$r = \frac{\sqrt{\text{Var}(X)}}{\sqrt{\text{Var}(Y)}} \hat{\beta}_1.$$

4 Numeric Example

Example: The Rocket Propellant Data

A rocket motor bonds an igniter propellant to a sustainer propellant inside a metal housing. The shear strength of that bond is a key quality characteristic, and it is suspected to depend on the age of the sustainer propellant batch.

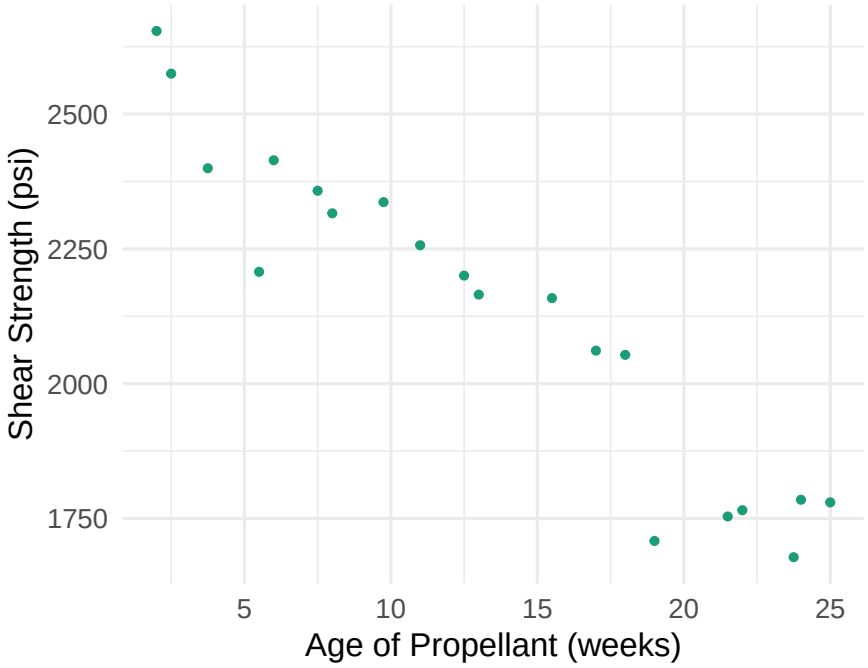
- $n = 20$ observations
- y_i: shear strength (psi)
- x_i: age of propellant batch (weeks)
- Tentative model: $y = \beta_0 + \beta_1 x + \varepsilon$

We will fit this model with OLS and check the algebraic properties from the previous frames against real numbers.

The Data ($n = 20$)

i	x_i (weeks)	y_i (psi)	i	x_i (weeks)	y_i (psi)
1	15.50	2158.70	11	13.00	2165.20
2	23.75	1678.15	12	3.75	2399.55
3	8.00	2316.00	13	25.00	1779.80
4	17.00	2061.30	14	9.75	2336.75
5	5.50	2207.50	15	22.00	1765.30
6	19.00	1708.30	16	18.00	2053.50
7	24.00	1784.70	17	6.00	2414.40
8	2.50	2575.00	18	12.50	2200.50
9	7.50	2357.90	19	2.00	2654.20
10	11.00	2256.70	20	21.50	1753.70

Scatter Diagram



Strength decreases as the propellant batch ages.

Computing S_{xx}, S_{xy}, and the Fitted Line

From the data: $n = 20$, $\bar{x} = 13.3625$, $\bar{y} = 2131.3575$.

$$S_{xx} = \sum_i x_i^2 - \frac{\left(\sum_i x_i\right)^2}{n} = 4677.69 - \frac{71{,}422.56}{20} = 1106.56$$

$$\begin{aligned} S_{xy} &= \sum_i x_i y_i - \frac{\left(\sum_i x_i\right)\left(\sum_i y_i\right)}{n} \\ &= 528{,}492.64 - \frac{(267.25)(42{,}627.15)}{20} = -41{,}112.65 \end{aligned}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41{,}112.65}{1106.56} = -37.15$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.3575 - (-37.15)(13.3625) = 2627.82.$$

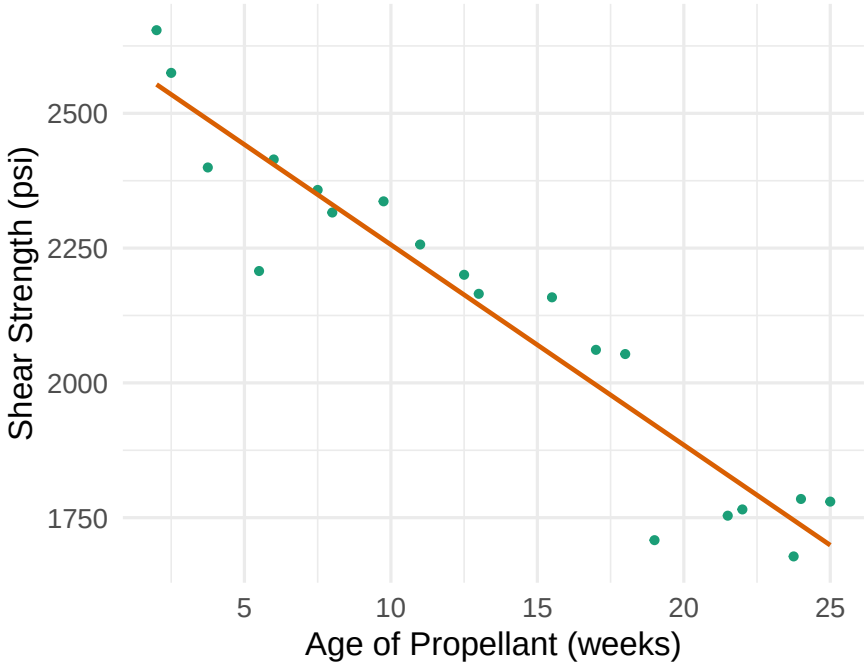
Fitted Equation

$$\boxed{\hat{y} = 2627.82 - 37.15x}$$

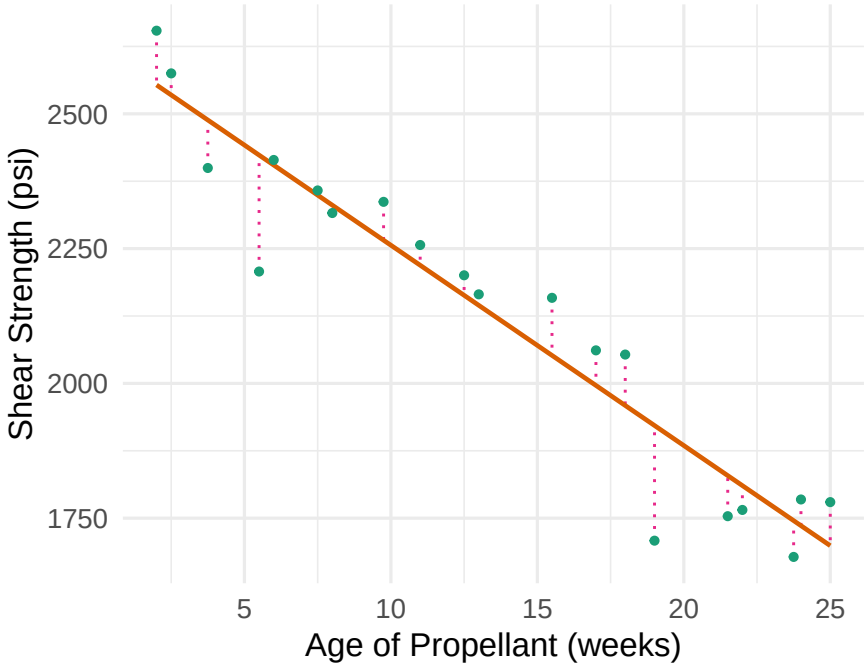
Each extra week of propellant age is associated with about 37 fewer psi of bond shear strength, on average.

What is the predicted value of the bond shear strength (Y) when the age of the propelland is 10 weeks?

Fitted Line



Fitted Line with Residuals



Fitted Values and Residuals

i	y_i	$\hat{y}_i$	$\hat{\varepsilon}_i$	i	y_i	$\hat{y}_i$	$\hat{\varepsilon}_i$
1	2158.70	2051.94	106.76	11	2165.20	2144.83	20.37
2	1678.15	1745.42	-67.27	12	2399.55	2488.50	-88.95
3	2316.00	2330.59	-14.59	13	1779.80	1698.98	80.82
4	2061.30	1996.21	65.09	14	2336.75	2265.58	71.17
5	2207.50	2423.48	-215.98	15	1765.30	1810.44	-45.14
6	1708.30	1921.90	-213.60	16	2053.50	1959.06	94.44
7	1784.70	1736.14	48.56	17	2414.40	2404.90	9.50
8	2575.00	2534.94	40.06	18	2200.50	2163.40	37.10
9	2357.90	2349.17	8.73	19	2654.20	2553.52	100.68
10	2256.70	2219.13	37.57	20	1753.70	1829.02	-75.32

Totals over all 20 observations: $\sum_i y_i = \sum_i \hat{y}_i = 42{,}627.15$, $\sum_i \hat{\varepsilon}_i = 0.00$.

5 Properties of the OLS Estimators

Rewriting $\hat{\beta}_1$ as a Linear Combination of the y_i

Since $\sum_i (x_i - \bar{x})\bar{y} = 0$, we can write

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_i (x_i - \bar{x}) y_i}{S_{xx}} = \sum_{i=1}^{n} c_i y_i, \qquad c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

The weights c_i satisfy two useful properties, both provable directly from the definition of S_{xx}:

$$\sum_i c_i = 0, \qquad \sum_i c_i x_i = 1, \qquad \sum_i c_i^2 = \frac{1}{S_{xx}}$$

These properties are needed to establish the unbiasedness of the estimators and find their variances.

Unbiasedness of $\hat{\beta}_1$

Treating the x_i as fixed and using $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

$$\mathrm{E}[\hat{\beta}_1] = \mathrm{E}\left[\sum_i c_i y_i\right] = \sum_i c_i \,\mathrm{E}[y_i] = \sum_i c_i(\beta_0 + \beta_1 x_i)$$

$$= \beta_0 \underbrace{\sum_i c_i}_{=0} + \beta_1 \underbrace{\sum_i c_i x_i}_{=1} = \beta_1$$

$$\boxed{\mathrm{E}[\hat{\beta}_1] = \beta_1}$$

On average, across repeated samples, the OLS slope estimate equals the true parameter.

A Preliminary Result: $\mathrm{E}[\bar{y}]$

Since $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\mathrm{E}[\varepsilon_i] = 0$:

$$\mathrm{E}[y_i] = \beta_0 + \beta_1 x_i$$

Averaging over $i = 1, \ldots, n$:

$$\mathrm{E}[\bar{y}] = \frac{1}{n}\sum_i \mathrm{E}[y_i] = \frac{1}{n}\left(n\beta_0 + \beta_1 \sum_i x_i\right) = \beta_0 + \beta_1 \bar{x}$$

Unbiasedness of $\hat{\beta}_0$

Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{E}[\bar{y}] = \beta_0 + \beta_1 \bar{x}$:

$$\mathrm{E}[\hat{\beta}_0] = \mathrm{E}[\bar{y}] - \bar{x}\;\mathrm{E}[\hat{\beta}_1] = (\beta_0 + \beta_1 \bar{x}) - \bar{x}\beta_1$$

$$\boxed{\mathrm{E}[\hat{\beta}_0] = \beta_0}$$

Both OLS estimators are unbiased.

Variance of the OLS Estimators

Because the ε_i are independent with common variance σ^2, and $\hat{\beta}_1 = \sum_i c_i y_i$ with $\mathrm{Var}(y_i) = \sigma^2$:

$$\mathrm{Var}(\hat{\beta}_1) = \sum_i c_i^2 \,\mathrm{Var}(y_i) = \sigma^2 \sum_i c_i^2 = \frac{\sigma^2}{S_{xx}}$$

A similar calculation for $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ gives

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$$

More spread-out x values (larger S_{xx}) give more precise slope estimates — intuitively, advertising spend that varies a lot from month to month tells us more about its effect on sales than spend that barely changes.

Variance of $\hat{\beta}_0$

Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$:

$$\begin{aligned}
\operatorname{Var}(\hat{\beta}_0) &= \operatorname{Var}(\bar{y}) + \bar{x}^2 \operatorname{Var}(\hat{\beta}_1) - 2 \operatorname{Cov}(\bar{y}, \hat{\beta}_1 \bar{x}) \\
&= \frac{\sigma^2}{n} + \bar{x}^2 \cdot \frac{\sigma^2}{S_{xx}} - 2\bar{x} \operatorname{Cov}\left(\bar{y}, \sum_{i=1}^{n} c_i y_i\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) - 2\bar{x} \sum_{i=1}^{n} c_i \operatorname{Cov}(\bar{y}, y_i)
\end{aligned}$$

$$\begin{aligned}
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) - 2\bar{x} \sum_{i=1}^{n} c_i \operatorname{Cov}\left(\frac{1}{n}\sum_{j=1}^{n} y_j,\ y_i\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) - 2\bar{x} \sum_{i=1}^{n} c_i \cdot \frac{\sigma^2}{n} \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) - 2\bar{x} \cdot 0 \cdot \sigma^2
\end{aligned}$$

$$\boxed{\operatorname{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}$$

The Gauss–Markov Theorem: OLS Is BLUE

Among all estimators of β_0 and β_1 that are:

- **L**inear in the y_i, and
- **U**nbiased,

the OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ have the smallest possible variance.

This is why OLS is called **BLUE**: the **B**est **L**inear **U**nbiased **E**stimator. "Best" here means minimum variance, not that OLS makes no errors — it means no other linear, unbiased method can do better, under the SLR assumptions.

6 Other Useful Properties of the Least-Squares Fit

Other Useful Properties of the Least-Squares Fit

Beyond unbiasedness and BLUE, the OLS fit has several algebraic properties that follow directly from the normal equations — worth knowing since they show up in diagnostics and in software output.

- The residuals always sum to zero: $\sum_i \hat{\varepsilon}_i = 0$
- The observed and fitted values have the same sum: $\sum_i y_i = \sum_i \hat{y}_i$
- The fitted line always passes through the centroid $(\bar{x}, \bar{y})$
- Residuals weighted by x_i sum to zero: $\sum_i x_i \hat{\varepsilon}_i = 0$
- Residuals weighted by $\hat{y}_i$ sum to zero: $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$

Property 1 & 2: Residuals Sum to Zero

Recall the first normal equation from earlier:

$$\sum_i y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

Since $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, summing over i gives

$$\sum_i \hat{y}_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

The right-hand sides are identical, so

$$\boxed{\sum_i y_i = \sum_i \hat{y}_i} \quad \Longrightarrow \quad \sum_i (y_i - \hat{y}_i) = \sum_i \hat{\varepsilon}_i = 0$$

This is not a coincidence — it is forced by the very equation OLS solves to find $\hat{\beta}_0$.

Property 3: The Fitted Line Passes Through $(\bar{x}, \bar{y})$

Divide the previous result $\sum_i y_i = \sum_i \hat{y}_i$ by n:

$$\bar{y} = \overline{\hat{y}} = \frac{1}{n} \sum_i (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

So the point $(\bar{x}, \bar{y})$ satisfies the fitted equation exactly — the **centroid** of the data always lies on the regression line.

This matches how we solved for $\hat{\beta}_0$ earlier: we built the line to pass through $(\bar{x}, \bar{y})$ by construction.

Property 4 & 5: Weighted Residuals Sum to Zero

Recall the second normal equation:

$$\sum_i x_i y_i = \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since $x_i \hat{y}_i = \hat{\beta}_0 x_i + \hat{\beta}_1 x_i^2$, summing gives $\sum_i x_i \hat{y}_i$ equal to the same right-hand side, so

$$\sum_i x_i y_i = \sum_i x_i \hat{y}_i \quad \Longrightarrow \quad \boxed{\sum_i x_i \hat{\varepsilon}_i = 0}$$

Property 4 & 5: Weighted Residuals Sum to Zero (cont.)

For the last property, write $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and use both results just derived:

$$\sum_i \hat{y}_i \hat{\varepsilon}_i = \hat{\beta}_0 \underbrace{\sum_i \hat{\varepsilon}_i}_{=0} + \hat{\beta}_1 \underbrace{\sum_i x_i \hat{\varepsilon}_i}_{=0} = 0$$

$$\boxed{\sum_i \hat{y}_i \hat{\varepsilon}_i = 0}$$

7 Estimating the Error Variance σ^2

Residuals and the Error Sum of Squares

The parameter σ^2 measures how much sales vary around the regression line for a given level of advertising spend — it is unknown and must be estimated from the residuals.

Define the **error sum of squares**:

$$\text{SS}_{\text{Res}} = \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

It can be shown that $\text{E}[\text{SS}_{\text{Res}}] = (n-2)\sigma^2$.

An Unbiased Estimator of σ^2

Since $\text{E}[\text{SS}_{\text{Res}}] = (n-2)\sigma^2$, we can write:

$$\boxed{\hat{\sigma}^2 = \text{MS}_{\text{Res}} = \frac{\text{SS}_{\text{Res}}}{n-2}}$$

The divisor $n-2$ reflects that two degrees of freedom were used up in estimating β_0 and β_1 before we could compute residuals.

Since $\text{E}[\text{SS}_{\text{Res}}] = (n-2)\sigma^2$, it follows that

$$\text{E}[\hat{\sigma}^2] = \frac{\text{E}[\text{SS}_{\text{Res}}]}{n-2} = \frac{(n-2)\sigma^2}{n-2} = \sigma^2$$

so MS_{Res} is an unbiased estimator of the error variance. This quantity is required for confidence intervals, and hypothesis tests for β_0 and β_1.

Appendix

Properties of the Weights c_i

Recall $c_i = \dfrac{x_i - \bar{x}}{S_{xx}}$. Each property follows directly.

$$\sum_i c_i = \frac{1}{S_{xx}} \sum_i (x_i - \bar{x}) = \frac{1}{S_{xx}} \left(\sum_i x_i - n\bar{x} \right) = 0$$

since $\sum_i x_i = n\bar{x}$ by definition of $\bar{x}$.

$$\sum_i c_i x_i = \sum_i c_i (x_i - \bar{x}) + \bar{x} \sum_i c_i = \frac{1}{S_{xx}} \sum_i (x_i - \bar{x})^2 + \bar{x} \cdot 0 = \frac{S_{xx}}{S_{xx}} = 1$$

$$\sum_i c_i^2 = \sum_i \frac{(x_i - \bar{x})^2}{S_{xx}^2} = \frac{1}{S_{xx}^2} \sum_i (x_i - \bar{x})^2 = \frac{S_{xx}}{S_{xx}^2} = \frac{1}{S_{xx}}$$

Finding the Expected Value of SS_{Res}

Why $\text{E}[\text{SS}_{\text{Res}}] = (n-2)\sigma^2$: Rewriting the Residual

Recall $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Substituting into $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$:

$$\hat{y}_i = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 x_i = \bar{y} + \hat{\beta}_1 (x_i - \bar{x})$$

So each residual splits into a piece explained by x and a leftover piece:

$$y_i - \hat{y}_i = (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})$$

Why $\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}] = (n-2)\sigma^2$ **(cont.)**

Squaring and summing over i:

$$\begin{aligned}\mathrm{SS}_{\mathrm{Res}} &= \sum_i \left[(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})\right]^2 \\ &= \underbrace{\sum_i (y_i - \bar{y})^2}_{S_{yy}} - 2\hat{\beta}_1 \underbrace{\sum_i (x_i - \bar{x})(y_i - \bar{y})}_{S_{xy}} + \hat{\beta}_1^2 \underbrace{\sum_i (x_i - \bar{x})^2}_{S_{xx}} \\ &= \mathrm{SS}_{\mathrm{Res}} = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx}\end{aligned}$$

Simplifying $\mathrm{SS}_{\mathrm{Res}}$

So far:

$$\mathrm{SS}_{\mathrm{Res}} = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx}$$

Since $\hat{\beta}_1 = S_{xy}/S_{xx}$, we have $S_{xy} = \hat{\beta}_1 S_{xx}$, so

$$\hat{\beta}_1^2 S_{xx} = \hat{\beta}_1(\hat{\beta}_1 S_{xx}) = \hat{\beta}_1 S_{xy}$$

The last two terms combine into one:

$$-2\hat{\beta}_1 S_{xy} + \hat{\beta}_1 S_{xy} = -\hat{\beta}_1 S_{xy}$$

$$\boxed{\mathrm{SS}_{\mathrm{Res}} = S_{yy} - \hat{\beta}_1 S_{xy}}$$

Expectation of S_{yy}

Subtracting $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}$ from $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

$$y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon})$$

Squaring and summing:

$$S_{yy} = \beta_1^2 S_{xx} + 2\beta_1 \sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + \sum_i (\varepsilon_i - \bar{\varepsilon})^2$$

The x_i are fixed constants and $\mathrm{E}[\varepsilon_i - \bar{\varepsilon}] = 0$, so the middle term vanishes in expectation. For the last term, using $\mathrm{E}[\varepsilon_i^2] = \sigma^2$ and $\mathrm{Var}(\bar{\varepsilon}) = \sigma^2/n$:

$$\sum_i (\varepsilon_i - \bar{\varepsilon})^2 = \sum_i \varepsilon_i^2 - n\bar{\varepsilon}^2 \;\Rightarrow\; \mathrm{E}\left[\sum_i (\varepsilon_i - \bar{\varepsilon})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$$

$$\boxed{\mathrm{E}[S_{yy}] = \beta_1^2 S_{xx} + (n-1)\sigma^2}$$

Expectation of $\hat{\beta}_1 S_{xy}$

Using $S_{xy} = \hat{\beta}_1 S_{xx}$ again:

$$\hat{\beta}_1 S_{xy} = \hat{\beta}_1^2 S_{xx}$$

Since $\mathrm{E}[\hat{\beta}_1] = \beta_1$ and $\mathrm{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$:

$$\mathrm{E}[\hat{\beta}_1^2] = \mathrm{Var}(\hat{\beta}_1) + \left(\mathrm{E}[\hat{\beta}_1]\right)^2 = \frac{\sigma^2}{S_{xx}} + \beta_1^2$$

$$\mathrm{E}[\hat{\beta}_1 S_{xy}] = S_{xx}\left(\frac{\sigma^2}{S_{xx}} + \beta_1^2\right) = \sigma^2 + \beta_1^2 S_{xx}$$

The Final Result

Combining with the previous page:

$$\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}] = \mathrm{E}[S_{yy}] - \mathrm{E}[\hat{\beta}_1 S_{xy}] = \left[\beta_1^2 S_{xx} + (n-1)\sigma^2\right] - \left[\sigma^2 + \beta_1^2 S_{xx}\right]$$

$$\boxed{\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}] = (n-2)\sigma^2}$$

The $\beta_1^2 S_{xx}$ terms cancel exactly — that cancellation is why $\mathrm{SS}_{\mathrm{Res}}$ isolates leftover error variance rather than variance explained by the true slope.